

Skin Signs of Systemic Disease

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Cutaneous Rheumatology

- There are a number systemic diseases that can affect the skin, hair, and nails, many are **AUTOIMMUNE** w/ tissue destruction in numerous organ systems
- Important to note
 - There are characteristic **changes of the connective tissue** which will aid in the **diagnosis of systemic conditions**
 - **diagnostic criteria** of each disease based upon **history, physical exam, and medical testing**

What is this?



Lupus Erythematosus:

acute to chronic (affecting only skin)

severity: benign vs severe, fatal, systemic

Classification

- Acute Cutaneous LE:
 - Practically always associated with systemic lupus erythematosus (**SLE**), highly **photosensitive**
- Subacute Cutaneous LE
 - Annular or papulosquamous
- Chronic Cutaneous LE
 - Hypertrophic or Verrucous LE
 - Lupus Panniculitis
 - Discoid
 - Oral Discoid LE

Systemic Lupus Erythematosus (**SLE**):

multisystem autoimmune disorder involving connective tissue & blood vessels.

DX SLE: 4/11 of RASH OR PAIN

1. **Malar/Butterfly Rash** - erythematous plaques, spares nasolabial folds
2. **Discoid Rash** - erythematous papules with adherent keratotic scaling and follicular plugging; atrophic scarring may **occur with chronic lesions**
3. **Arthritis** - **pain** in 2 or more joints
4. **Serositis** - **fluid** around the heart or lung
5. **Hematologic disorder** - hemolytic anemia (RBC), leukopenia < 4000/uL (WBC), lymphopenia < 1500/uL , thrombocytopenia < 100,000/uL (Platelets)
6. **Oral ulcers** - painless
7. **Renal disease** - proteinuria (protein in urine)
8. **Photosensitivity**
9. **Antinuclear antibody (ANA):** >95%
10. **Immunologic disorder: autoantibodies (next slide)**
11. **Neurologic disorder** - seizures or psychosis

Autoantibodies in SLE

- **anti-dsDNA** (can indicate kidney and skin disease)
- **anti-Smith** (kidney disease)
- anti-nucleosomes (kidney and skin disease)
- **anti-Ro/SS-A** (skin and kidney disease, fetal heart problems)
- anti-La/SS-B (fetal heart problems)
- **anti-histone** (drug-induced lupus most commonly caused by “HIP” - Hydralazine, Isoniazid, and Procainamide)
- **anti-phospholipid** (thrombosis, pregnancy loss)
- anti-NMDA receptor (brain disease)
- anti- α -actinin (kidney disease)
- anti-C1q (kidney disease)
- VDRL (false positive for syphilis)

Subacute Cutaneous LE

- Annular – erythematous rings
- Papulosquamous - psoriasiform
- 50% develop 4 or more criteria for Systemic Lupus Erythematoses (SLE)
- 60-80% have a positive ANA
- **Anti-Ro (SS-A) positive in > 80%**, anti-La (SS-B) positive in 30-50%

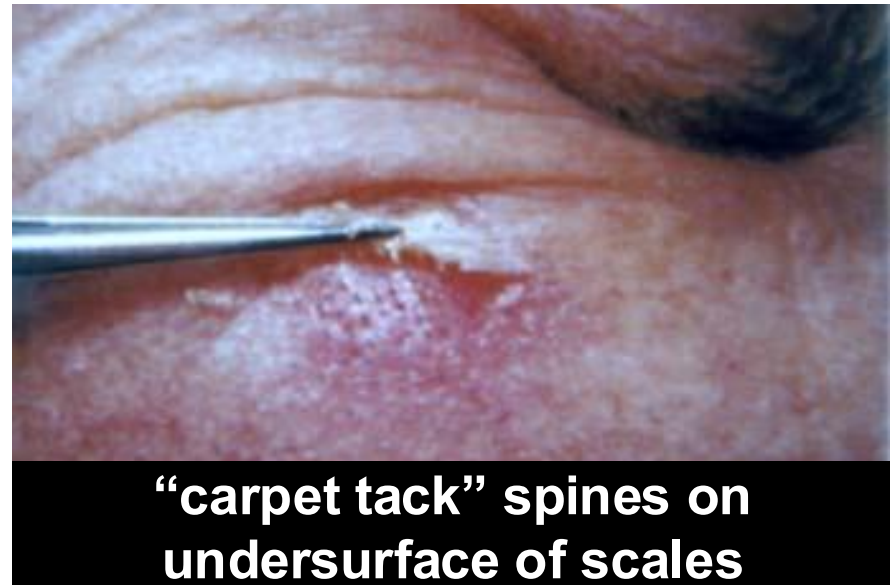


Chronic Cutaneous LE - Hypertrophic or Verrucous



CHRONIC CUTANEOUS LE

Discoid Lupus Erythematoses



Chronic Cutaneous LE - Discoid vs Oral Discoid



Discoid

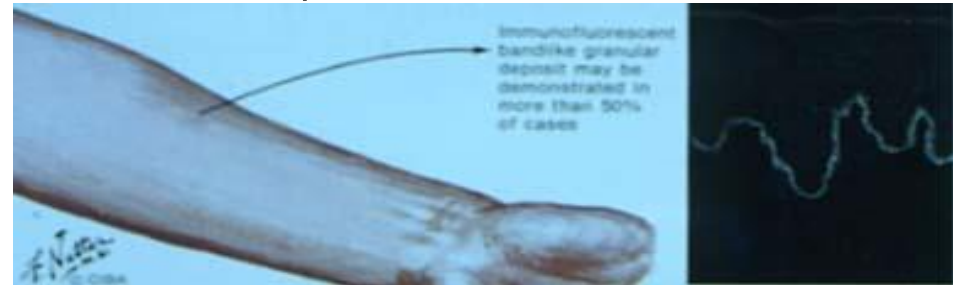
round, regressing in center, atrophy



Oral discoid

Evaluation & Tx of Patients With Lupus

- History and Physical
- Bloodwork: CBC, autoantibodies
- Urinalysis - proteinuria, hematuria
- Biopsy - lesional, non-lesional, and non-sunexposed
 - Immunofluorescence (**lupus band test**)
 - Positive in lesional skin 90%
 - Normal sun exposed 75%
- Tx:
 - Sunscreens
 - Steroids (Topical, Intralesional, Systemic)
 - anti-malarials (i.e. hydroxychloroquine)
 - Immunosuppressants



Dermatomyositis

Autoimmune disease targeting the skin and muscle

DX: 2/4 muscle findings PLUS CUTANEOUS finding = probable

- 1. Proximal symmetrical muscle weakness** involving trunk, shoulders, neck (i.e. difficulty rising from chair)
- 2. Elevated muscle enzymes - CPK, aldolase, LDH, transaminase**
- 3. Abnormal EMG**
- 4. Abnormal muscle biopsy**

Cutaneous:

- **Heliotrope rash****
- **Gottron's papules****
- Periungual telangiectasias
- Poikiloderma
- Malar erythema
- Photosensitivity

**** PATHOGNOMONIC**

Etiology: drugs, infections, trauma, vaccination, malignancy, pregnancy, abnormal circulating antinuclear ab (anto-Jo-1)



Heliotrope Rash



**Gottron's Papules
(over bony prominences)**

Source: IMACS

Periungual Telangiectasias and Cuticular Hypertrophy



Poikiloderma: erythema, pigmentation, telangiectasias, atrophy



Systemic Manifestations

- AM stiffness of joints
- Esophageal dysphagia - difficulty swallowing
- Lung disease - associated with anti-Jo-1
- Heart disease - cardiomyopathy
- **Malignancies** - mc. **ovarian cancer** in females.

Evaluation & Tx of Dermatomyositis

- History - malignancy, infections, drugs
- Physical and testing
 - **Female: pelvic (pap smear), breast exam (mammo)**
 - **Male: rectal exam**
 - **Both-chest x-ray, stool guiac (test blood in stool)**
- Evaluation of muscle disease: muscle biopsy, EMG
 - PFT's (pulmonary fxn tests), echocardiogram
 - Esophageal studies-muscle weakness
- Labs
 - **Creatine phosphokinase (CPK** - most specific for muscle disease)
 - **Antibody studies (anti-Jo-1)**

Tx: Physical therapy (helps muscles), prednisone (systemic steroids), immunosuppressants

Scleroderma - “sclerosis of skin”

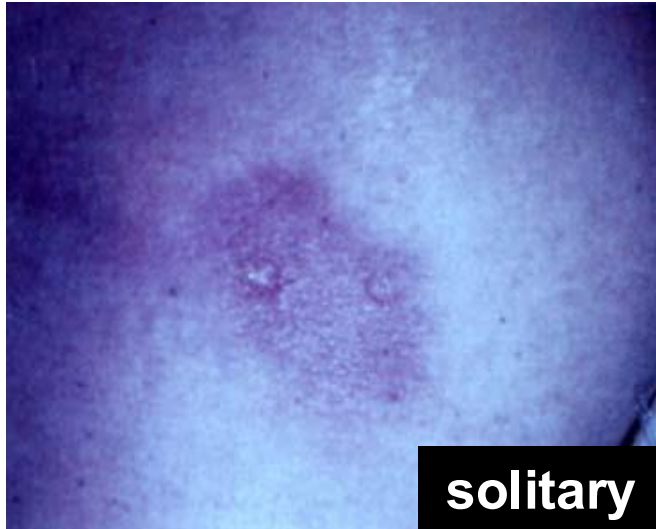
- Multi-system **autoimmune disease**
- **Fibroblasts overproduce collagen → fibrosis**
- Sclerotic changes of skin and internal organs (lungs, heart, GI)
- 2 Categories:
 - Localized: morphea
 - Systemic:
 - limited (CREST vs acrosclerosis)
 - progressive (multi system)

Localized Scleroderma - Morphea

- Violaceous to ivory hardened skin, with possible underlying atrophy (depression)
- It may be:
 - Solitary
 - Generalized
 - Linear
 - “**En coup de sabre**” - linear morphea of the head
 - Pansclerotic: involvement of dermis, fat, muscle, bone (disabling)

Localized Scleroderma – Morphea

Solitary plaque vs Generalized vs Pansclerotic



Localized Scleroderma - Morphea
**Linear type, En coup de sabre (strike of
saber)**



Systemic Scleroderma

Limited Type: Acrosclerosis

- Limited to hands, feet, face and arms



Systemic Scleroderma

Limited Type: CREST

- **Calcinosis:** calcium deposits in skin, anti-**C**entromere ab
- **R**aynaud's
- **E**sophageal dysfunction
- **S**clerodactyly (thick/tight tapered skin of the fingers and toes)
- **T**elangiectasias (hands, face)



Systemic Scleroderma

Progressive Type (PSS): skin & internal organs

- Raynaud's (1st manifestations of PSS)
- Pulmonary **fibrosis**
- Kidney failure
- Heart attack and failure
- “mask like facies”, loss of facial lines



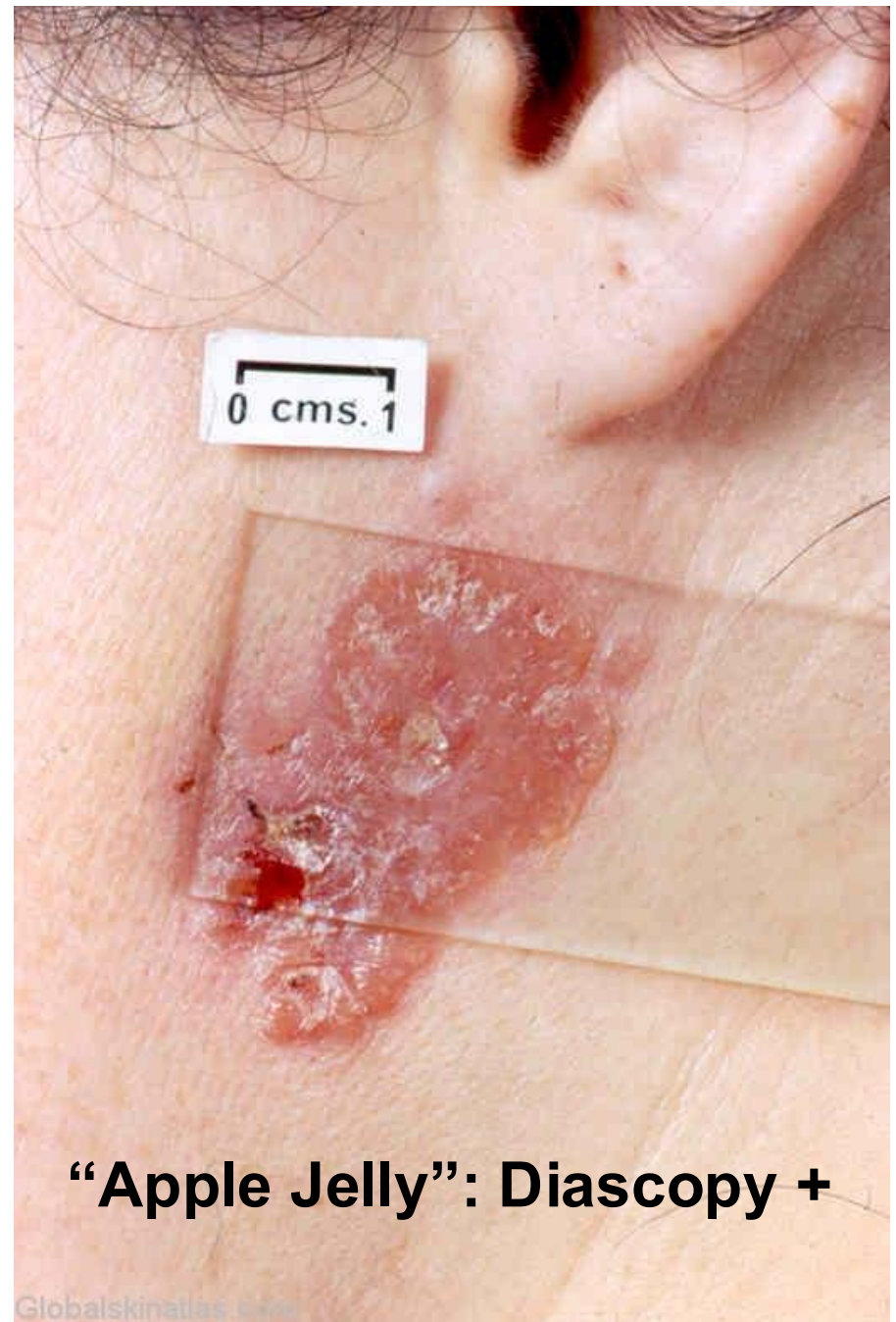
Diagnosis and Treatment

- Blood tests
 - **ANA**
 - **anti-Scl70**
 - **anti-centromere (CREST)**
- Tests for condition affecting internal organs
- Treatment: glucocorticoids, immunosuppressants

Sarcoidosis

- Systemic **granulomatous disease** of unknown etiology; more common in **African Americans**
- Primarily affects the **lungs** - bilateral hilar lymphadenopathy
- **Skin** - **granulomatous** lesions, **lupus pernio**, brown-purple papules
 - lesions are **diascopy positive**: blanching with a glass slide reveals “**apple-jelly**” **color** in the lesions
 - **Erythema nodosum** is the most common non-specific lesion in the skin (good prognosis)
- **non-caseating granulomas**
- **hypercalcemia**
- elevated angiotensin-converting enzyme (**ACE**)

Granulomatous Lesions



“Apple Jelly”: Diascopy +

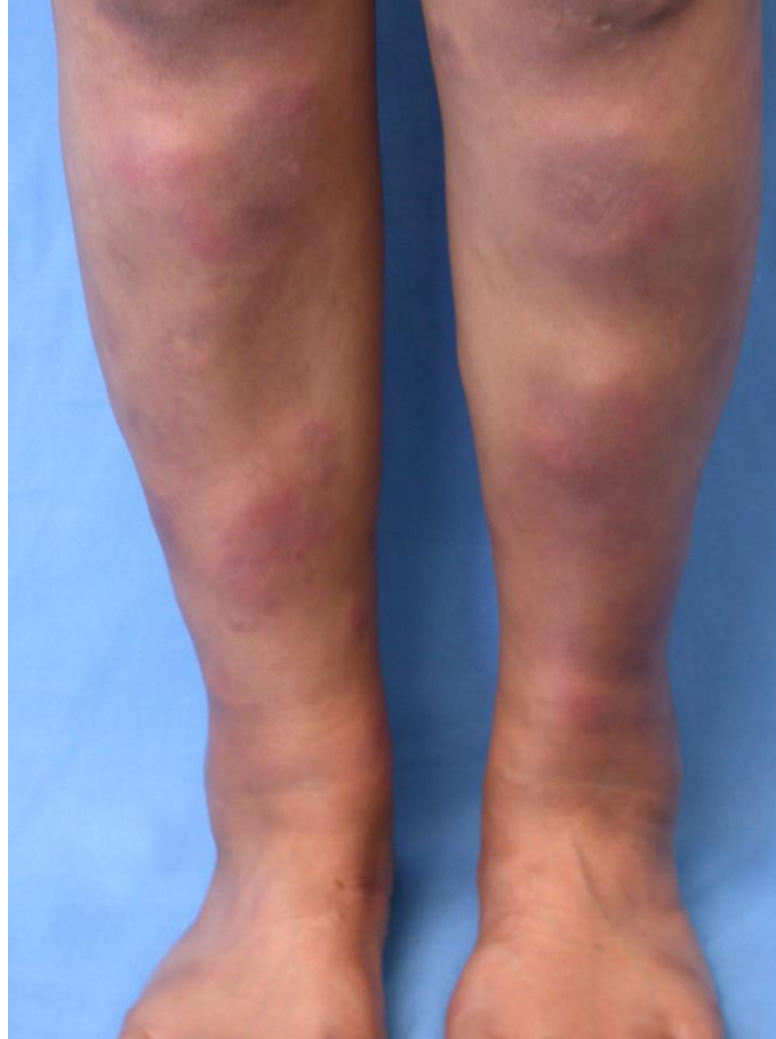
Globalskinatlas.com



Lupus Pernio

Sarcoidosis - Erythema Nodosum

nonspecific finding, best prognosis



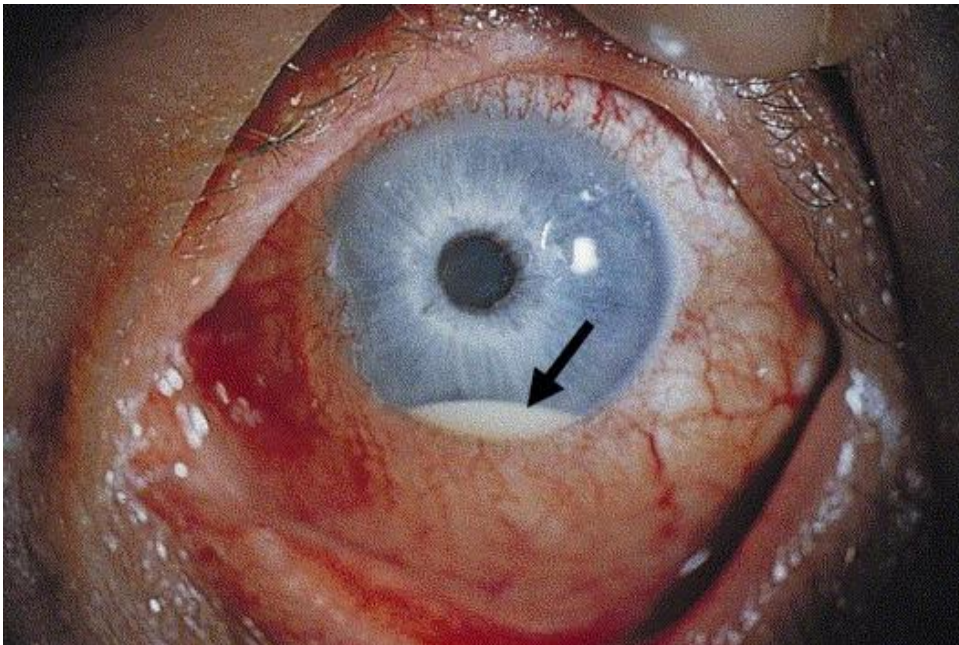
Behçet's Disease

- **Recurrent oral aphthous ulcers**
- **2 of the following:**
 - **Genital aphthosis:** *commonly misdiagnosed as herpes simplex virus (HSV)*
 - Uveitis
 - Arthritis (non-erosive)
 - Skin lesions: cutaneous pustular vasculitis
 - Meningoencephalitis
- Diagnosis - **Pathergy test** (inflammatory pustule found 24-48 hours after the skin is punctured by a needle)

Oral Aphthosis



Genital Aphthosis



← Anterior Uveitis
(can be anterior or posterior)

Kawasaki Disease aka

Mucocutaneous Lymph Node Syndrome

5/6 Criteria: CRASH & Burn

- **Conjunctival Injection**
- **Rash:** erythematous, spreads from palms/soles to trunk
- **Adenopathy**, lymph node mass $> 1.5\text{cm}$
- **Strawberry tongue**, oral fissures, lip crusting
- **Hand/foot edema**, erythema, induration, desquamation of digits
- **Burn:** unexplained fever $> 5\text{days}$

Affects children $< 10\text{yo}$, 2% mortality: ruptured coronary aneurysm

Tx: salicylates, IVIG



Case 1: Sun Kissed

- Ms. F (32 yo childcare employee) c/o easily fatigued arms and difficulty lifting children.
- rash on cheeks, across nasal bridge, around eyes after winter vacation in Barbados.
 - Pain in and blanching of index, middle, and ring fingers after shoveling snow but resolves after 20 minutes of warmth.
 - PMHx of hypercholesterolemia, currently on atorvastatin.
 - FamHx: Mother (deceased): autoimmune disorder

Notable PE Findings:

MSK: Bilateral upper and lower extremity weakness (strength 2/5)

Derm: erythematous to hyperpigmented rash to the posterolateral aspect of the neck, telangiectasia, rash over elbows and MCP, PIP, DIP joints.

What is your differential diagnosis?

Which labs and diagnostic tests would you perform?

Case 1: Sunkissed

	Likely due to:	Unlikely due to:	Tests:
Dermatomyositis	Rash over bony prominences Photosensitivity Rash around eyes Bilateral upper and lower extremity weakness Raynaud's		• CPK, aldolase, LDH, transaminase • EMG
SLE	Erythematous malar rash Photosensitivity Raynaud's	Muscle findings	• Auto-antibodies • Antinuclear ab's • Lupus band test (rule out discoid lupus) • Urinalysis?
Sarcoidosis	Erythematous rash across nasal bridge and cheeks	Lungs clear	• Diascopy • Calcium levels • ACE levels
RA/OA	MCP, PIP, DIP joints.	Overlap of involved joints	• Rheumatoid factor • Anti-CCP • XRay
Scleroderma	Raynaud's	Lack of fibrosis, plaques, other organ involvement	• ANA • anti-Scl70 • anti-centromere (CREST)

Case 1: Labs and Tests

-CBC

-Antibody testing (ANA, Anti-Jo1)

- What if ANA was negative, Anti-Jo1 was positive?

-Muscle enzyme panel

- What if CPK, LDH, transaminases were elevated?

-Muscle biopsy

-EMG

Case 2: 'tis the Season

Mr. Smith (30 yo landscaper) c/o mouth sores, shortness of breath, photophobia, eye redness and pain, knee pain, skin discoloration on leg and genitalia.

- Denies floaters, visual loss
- Believes work stress is causing a new seasonal allergy and red eyes. No relief from OTC antihistamines or avoiding the outdoors.
- Knee pain not relieved by knee pads.
- No sexual activity for 1yr after a long term monogamous relationship.

Notable PE findings:

Eyes: hypopyon, 360 degree injection around the cornea

Mouth: aphthous ulcers

Extremities: tenderness and erythema on bilateral anterior lower extremities

GU: aphthous ulcers

What is your differential diagnosis?

Which labs and diagnostic tests would you perform?

Case 2: 'tis the Season

- **CBC:** normal except for a mildly elevated reticulocyte count.
- **ESR** and **CRP:** elevated (something inflammatory (vasculitis involvement?), could be lupus because of the arthritic pain as well)
- **CMP:** normal calcium levels (decreasing likelihood of sarcoidosis)
- **Serum Angiotensin converting enzymes-** were normal (decreasing likelihood of sarcoidosis)
- **Serum antibody levels** are negative for **ANA** (ruling out SLE) and **anti-Jo** (ruling out dermatomyositis).
- **CXR** showed opacities and nodular shadows with the beginnings of a pleural effusion
- **Sputum analysis** (grossly blood tinged) showed RBCs, but no acid fast bacilli. (lowering TB on differential list)
- **Tuberculin testing** was negative.

Why are anti scl70 and anti-centromere unjustified tests?

- Absence of calcinosis, raynaud's, sclerodactyly, renal or cardiac involvement

Which dermatological findings are shared between Behcet's and Sarcoidosis?

- Granulomatous lesions

- Erythema Nodosum**

- “Apple Jelly” lesions

- Lupus pernio

- Morphea

- Meningoencephalitis

Case 3: Sunbathing for Brad

Ms. Robbins (61 yo librarian) c/o **lethargy**, pale skin, “sun allergy”, and ugly hand joints

–Went sunbathing because she didn’t want to be so **pale** when she saw her son, **Brad Pitt for Mother’s Day (delusional)**. Discovered a “**sun allergy**” due to **red rash** on cheeks

–Current medications: **multi vitamin**

–Denies drug, smoking, ETOH (**psychosis in absence of psych altering substances**)

Notable PE findings:

Face: **raised erythematous rash sparing nasolabial folds** → malar rash?

Eyes: **conjunctival pallor** + lethargy → anemia?

Pulm: CTA bilaterally

Extremities: swollen DIP, PIP → **arthritis**?

What is the your differential?

Case 3: Sunbathing for brad

	Likely due to:	Unlikely due to:	Tests:
SLE	Erythematous malar rash Anemia Photosensitivity Delusions Arthritic findings RASH OR PAIN		• Auto-antibodies • Antinuclear ab's • Lupus band test (rule out discoid lupus) • Urinalysis?
Dermatomyositis	Photosensitivity Malar erythema	Lack of muscle findings	• CPK, aldolase, LDH, transaminase
Sarcoidosis	Erythematous rash	Spares nasolabial folds Lungs clear	• Diascopy • Calcium levels
Rheumatoid Arthritis	PIP swelling	DIPs usually spared	• Rheumatoid factor • Anti-CCP

Human Graft vs. Host Disease (GVHD)

- Graft with **immunocompetent donor** cells
- **Immuno-incompetent** recipient
 - organ is rejected by the host
- **Histoincompatibility**
- Types: (differ in pathogenesis)
 - **Acute**: 14-100 days post-BMT
 - **Chronic**: > 100 days post-BMT

Acute Graft v Host –

14-100 days post-BMT
affecting skin, liver, and GI



Classic maculopapular rash

Acute Graft v Host

14-100 days post-BMT
affecting skin, liver, and GI



TEN-like (Toxic Epidermal Necrolysis)

Severe cases: maculopapular rash → forms bullous lesions with TEN
(mimics Stevens-Johnson syndrome)

Chronic Graft v Host

>100 days post-BMT
affecting many organs



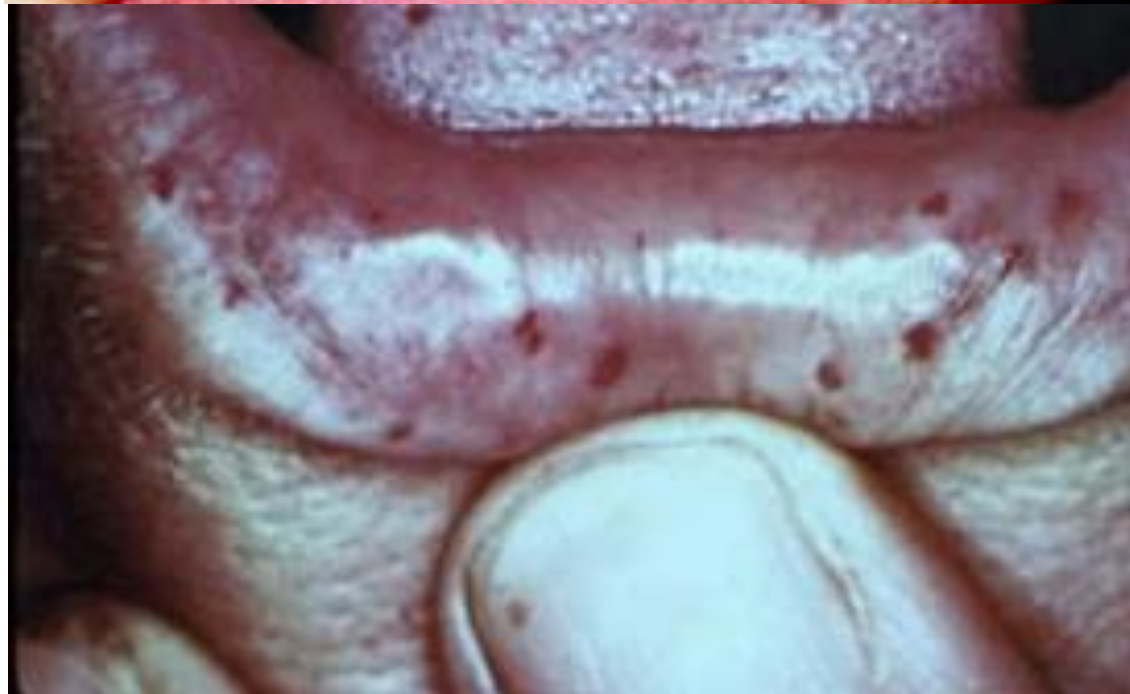
skin sclerosis

Cutaneous disease and gastrointestinal abnormalities

Hereditary Hemorrhagic Telangiectasia (HHT) (Osler-Weber-Rendu Syndrome)

- Autosomal dominant
- **Telangiectasias** of the **mouth** and **GI tract**
- Early sign of disease: **epistaxis**
- Complications:
 - **melenas** (bleeding bowel)
 - **epistaxis** (bleeding nose)

HHT



Superficial telangiectasias of the mouth

spontaneous and recurrent
nosebleeds (epitaxis)

multiple mucotaneous
telangiectasia

Diagnostic Criteria for HHT

visceral involvement (GI,
pulmonary, cerebral, or
hepatic AVMs)

first degree relative with
HHT

Gardner Syndrome

- Autosomal dominant
- Adenomatous **colorectal polyposis** (> 100 polyps) -**malignant potential**
- Cutaneous findings:
 - **Multiple epidermal cysts**: face, scalp (often)
 - **Osteomas**: mc of the jaw
 - **Dental abnormalities**: supernumerary teeth

Gardner Syndrome—cutaneous findings

Multiple epidermal cysts

- Most commonly found on the **face**
- Often occur before puberty and precede the onset of polyposis



Gardner Syndrome—cutaneous findings

Osteomas

- Most commonly of the **jaw**



Gardner Syndrome—cutaneous findings

Dental abnormalities

- **Supernumerary teeth**, unerupted teeth



Non-Malignant Polyposis Syndromes

- **Peutz-Jeghers Syndrome**
- **Cowden's Syndrome aka Multiple Hamartoma Syndrome**

Peutz-Jeghers Syndrome

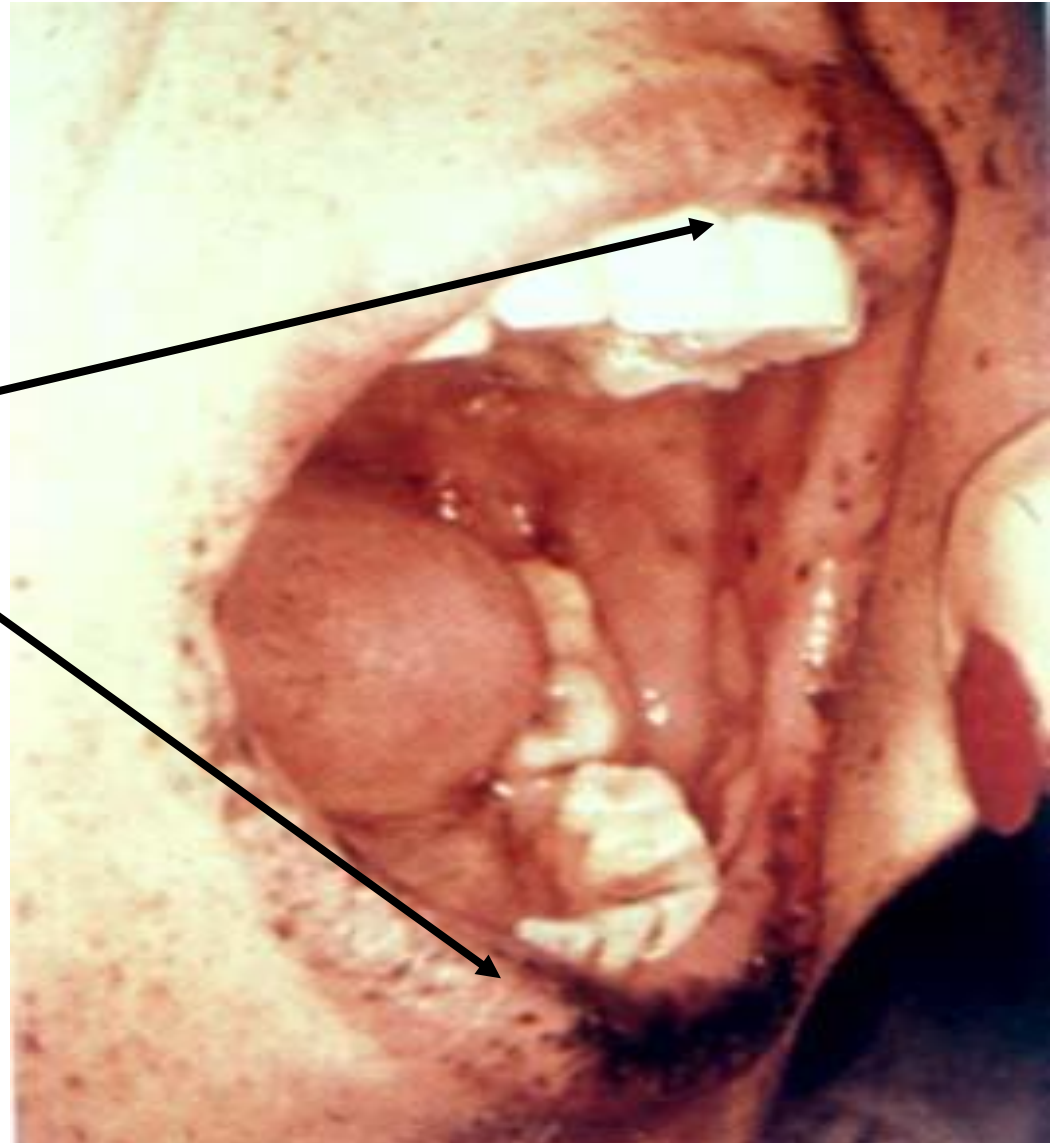
- Autosomal dominant
- Multiple hamartomatous **GI polyps**
- **Spotty melanin pigmentation**
- Increased risk of GI and non-GI cancer

Peutz-Jeghers Syndrome

non-malignant GI polyps, pigmentation of the lips

Spotty melanin pigmentation:

- Mostly found on the **lips** and perioral region



Cowden's Syndrome aka Multiple Hamartoma Syndrome

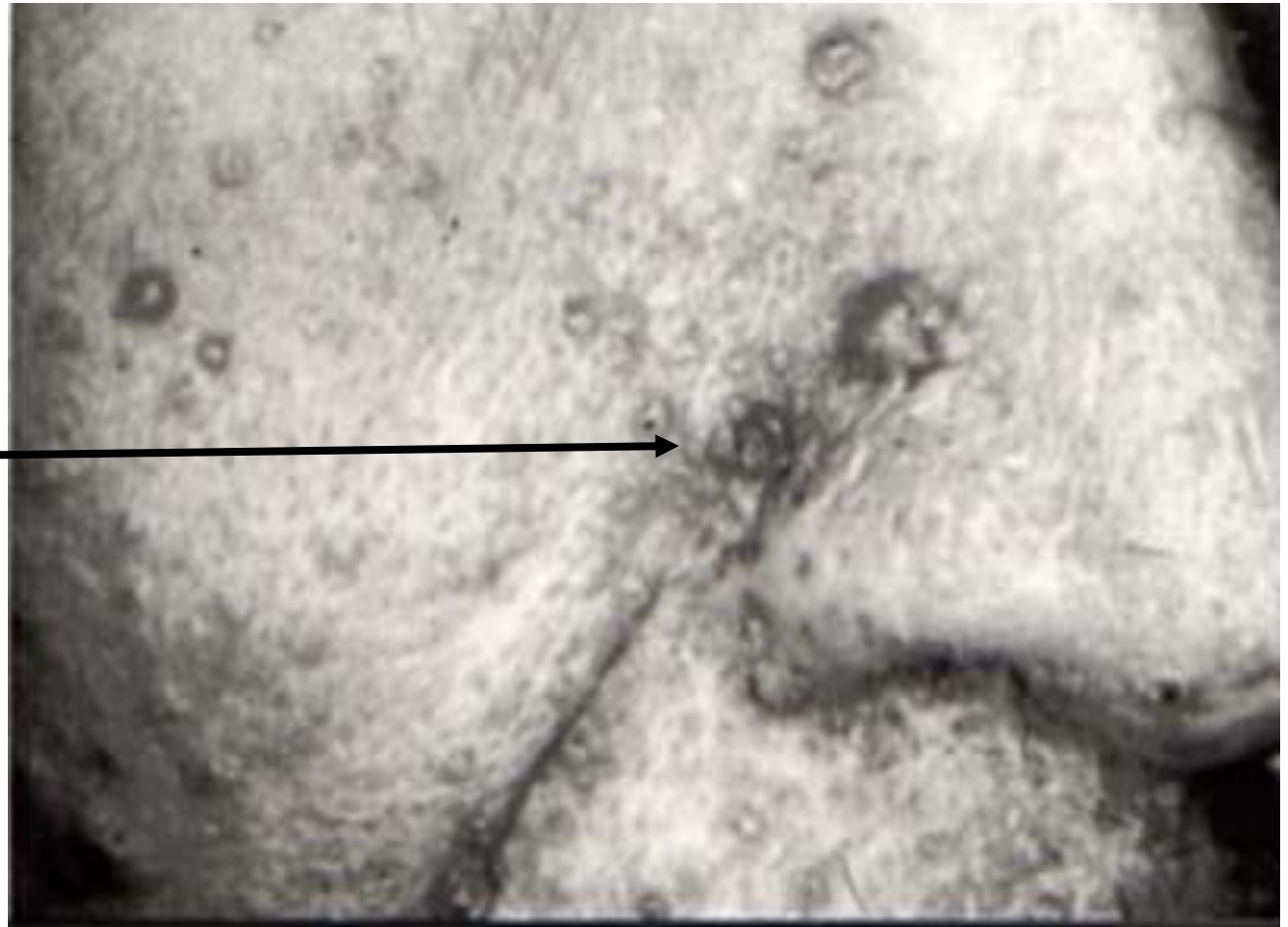
- Autosomal dominant
- **GI polyps**
- **Warty papules**
- Multiple hamartomatous tumors in multiple organ systems
- Increased risk of malignancy- breast Ca

Cowden's Syndrome

non-malignant GI polyps, warty papules

Warty papules

- Found on the **face** and **forearms**



Acrochordons

“Skin tags”

- Commonly found on the **neck, axilla,** and **groin** are found to be **associated with colonic polyps**
- Skin tags: outgrowths of normal skin in sites of friction
 - 50% of adults, risk increases with age
 - associated with obesity and DM
 - Tx: removal with forceps and scissors

Acrochordons



on axilla



on neck

Skin Signs Associated with Certain Hepatologic Conditions

Cirrhosis of the Liver

- Patients frequently develop skin jaundice and **spider telangiectasia (angiomata)**
- **Itching of the skin**
- **Palmar erythema, nail changes (Terry nails)**

Cirrhosis of the Liver- Cutaneous Findings

Spider telangiectasias

- Vascular lesions consisting of a central arteriole surrounded by smaller vessels
- Most frequently on the **face**, trunk, and upper limbs



Spider
telangiectasias
on the trunk
and upper limbs



Terry nails

- Proximal 2/3rd of nail plate appears white
- Non-specific for liver disease



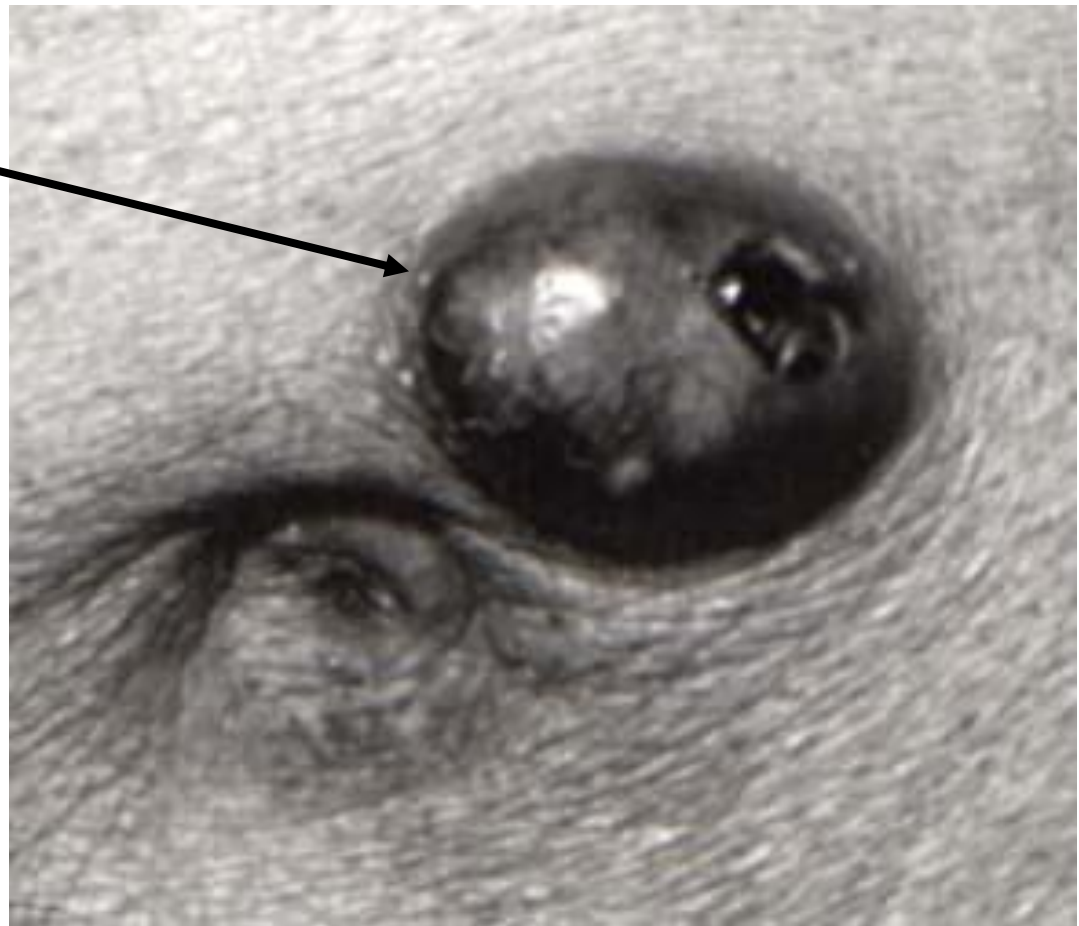
Pancreatic Cancer and Skin Disease

Sister Mary Joseph's Nodule

- Malignant metastatic umbilical nodules (Sister Mary Joseph's Nodule)

Sister Mary Joseph Nodule

- May be the first sign of pancreatic cancer
- Majority of patients with this nodule have a **poor prognosis**



Skin Signs of Certain Cardiovascular Conditions

Relapsing Polychondritis (RPC)

- Immune mediated
- **Recurrent inflammation affecting cartilage**
- Initial presentation: **tender, erythematous, warm, swollen ear.**
- Joint involvement
 - Arthritis seen in up to 1/3rd of patients

RPC

Recurrent inflammation of cartilage

- Particularly **ears**, nose, eyes, joints, and respiratory tract



Relapsing Polychondritis & Cardiovascular conditions

- **May occur with heart-aortic insufficiency and aortic aneurysms**
- **Occurs with aortic or mitral valvular disease in ~10% of patients**

Skin Signs of Certain Renal Conditions

Chronic Renal Failure related Skin Findings

- **Anemia-pale complexion**
- **Generalized hyperpigmentation**
- **Xerosis** (dry skin)
- **Pruritus** (itchy skin)
- **Uremic frost** (urea crystals form because waste products can't be excreted cause of kidney failure)
- **Metastatic calcification in skin**
- **Bullous Disease of Dialysis**



<http://emedicine.medscape.com/article/1094846-overview>

(bullous disease of dialysis)



Clin Kidney J (2014) 0: 1–2
doi: 10.1093/ckj/sfu057 (uremic frost)

Skin Signs of Malnutrition and Vitamin Deficiencies

Kwashiorkor-Protein Malnutrition

Major Cutaneous findings:

- **peripheral pitting edema**
- Peeling “**enamel paint**” dermatoses
- Depigmented bands in hair (**FLAG SIGNS**)

Kwashiorkor-Protein Malnutrition

Other clinical findings:

- Distended abdomen
- “Moon face”-Rounded prominence of cheeks
- Muscle atrophy
- Bradycardia, hypotension, and hypothermia

Kwashiorkor-Protein Malnutrition

Symmetric peripheral pitting edema:

- Mild involves only feet
- Moderate involves feet, legs and/or upper limbs
- Severe generalized edema



Kwashiorkor-Protein Malnutrition

Peeling “Enamel Paint” dermatoses

- Thin, dry, peeling skin with areas of **hyperkeratosis** and **hyperpigmentation**



Kwashiorkor-Protein Malnutrition

FLAG SIGN

- de-pigmented bands in hair
- Intermittent periods of adequate dietary intake restores hair color → results in alternating bands of normally colored and abnormally discolored hair



Acrodermatitis enteropathica

- **Deficiency of zinc due to malabsorption**
 - Due to autosomal recessive partial **defect in intestinal zinc absorption**
- Occurs in **infants**
- Common findings:
 - **Alopecia, diarrhea, severe growth retardation, frequent infections**, ophthalmic disorders, delayed sexual maturation, neuropsychiatric manifestations
- Tx: **zinc sulfate**

Acrodermatitis enteropathica

Cutaneous findings:

- **Vesiculobullous and erythematous dermatitis**
 - may look like atopic dermatitis
 - **acral, perioral, and perirectal areas are most common**



Acrodermatitis enteropathica



Figure 1: Acrodermatitis Enteropathica on 15 month old female characterised by periorificial and acral dermatitis and alopecia. There are annular, well demarcated, erythematous, crusted plaques and vesicles distributed symmetrically around orifices and hands.

Dermatitis Herpetiformis- Celiac Disease/Gluten Sensitive Enteropathy

- Autoimmune insensitivity to **gluten** (wheat, barley and rye) → **cutaneous eruption**
- **Autoantibodies**: anti-gliadin, endomysial, reticulin, and tissue transglutaminase
- Skin biopsy

Dermatitis Herpetiformis- Celiac Disease/Gluten Sensitive Enteropathy

Cutaneous Findings:

- intensely **pruritic inflammatory grouped papules and vesicles** found on extensor surfaces (**elbows, forearms, knees, scalp, or buttocks**)
- Treatment: Dapsone, **gluten-free diet**; typically responds well to treatment

Celiac Disease - Dermatitis Herpetiformis

Cutaneous Findings



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Grouped papules and vesicles on forearms and elbows

Celiac Disease - Dermatitis Herpetiformis

Cutaneous Findings



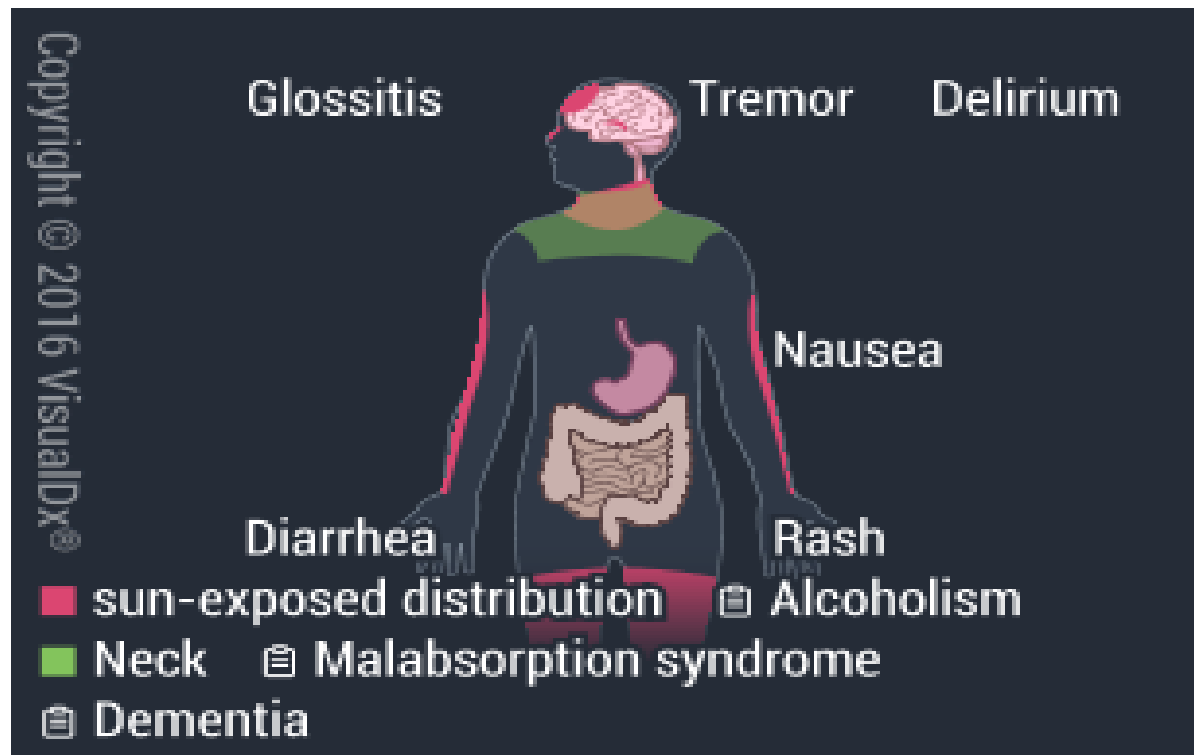
On knee



On buttocks

Niacin-Pellagra

- **Niacin deficiency** (vitamin B3)
- Pellagra with 3 D's
 - Dermatitis, Diarrhea, Dementia



Niacin-Pellagra

- **Dermatitis**:
erythematous
eruption in **sun
exposed** distribution
 - **CASAL'S
NECKLACE**
(dermatitis
around the neck)



Niacin-Pellagra

Dermatitis (Casal's Necklace)

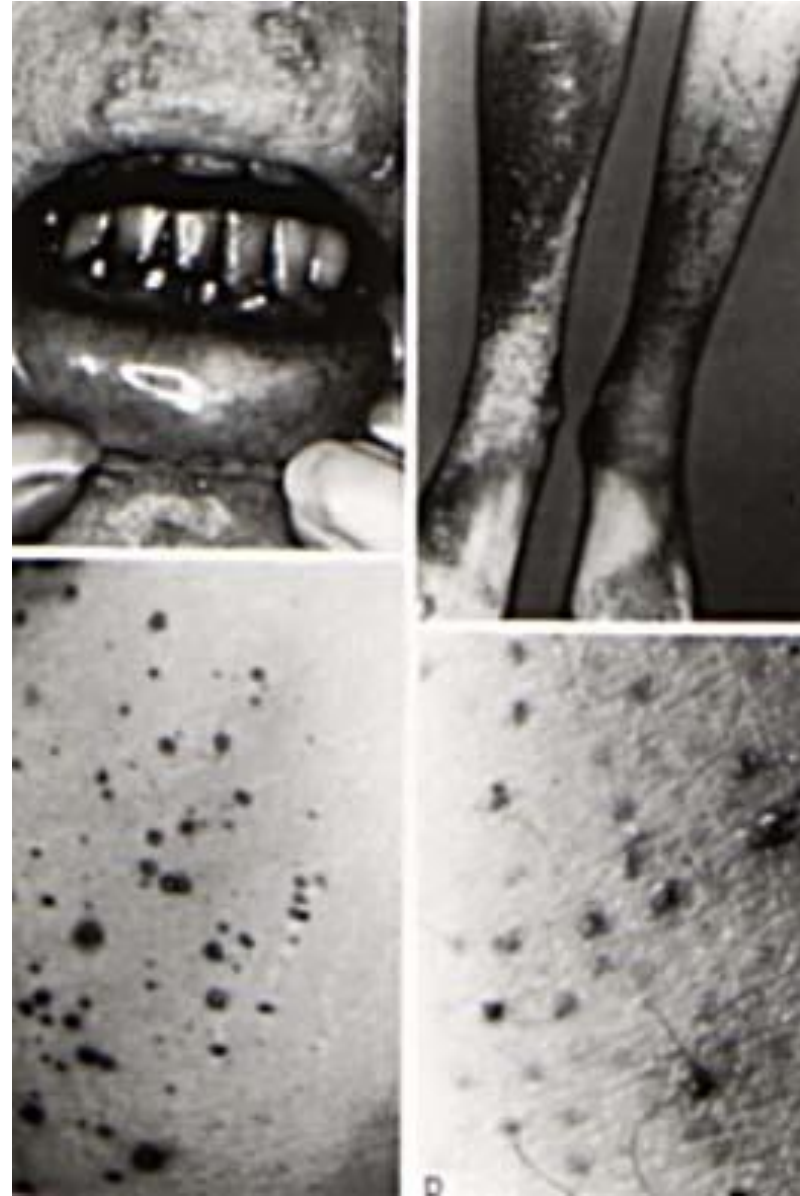


Scurvy- Vitamin C Deficiency

vitamin C deficiency causes sCurvy due to a Collagen synthesis defect

Cutaneous Findings:

- **Follicular hyperkeratotic papules**
- **“corkscrew hairs”**
- **gingivitis, swollen gums**
- **slow wound healing**
- **woody edema of legs**
- **nodules, nosebleeds, hematuria**



Scurvy

Follicular hyperkeratotic papules

- Reddish/bluish bruise like spots surrounding hair follicles



Review & Mnemonics

- Lupus Erythematosus: RASH OR PAIN
- Dermatomyositis: Heliotrope rash & Gottron's papules
- Kawasaki Disease: CRASH & Burn
- Behcet's Disease: recurrent oral & genital lesions, uveitis
- Sarcoidosis: Lung infiltrates & apple jelly skin
- Scleroderma: Localized vs Systemic
- Human Graft vs Host Disease: attacking foreign tissue, acute vs chronic
- HHT/OWR: nose and GI bleed
- PJS: GI polyps lip spots
- Cowdens/MHS: multiple tumors, facial warts
- Gardner: bumpy face and jaw with too many teeth, GI polyps
- Acrochordons: skin tags associated w/ DM, obesity, chronic G polyps
- Cirrhosis of liver: jaundice, spider telangiectasias, palmar erythema, terry nails
- Pancreatic cancer: Sister Mary Joseph nodule
- CV: relapsing polychondritis – painful cartilage
- Chronic renal failure: anemia, xerosis, pruritis, uremic frost, calcification
- Dermatitis Herpetiformis (Celiac Dx / Gluten): pruritus, grouped papules/vesicles on extensors
- Kwashiorkor: edematous malnutrition with flag sign and enamel paint
- Acrodermatitis enteropathica: baby w/ zinc deficiency, vesiculobullous and erythematous, diarrhea, alopecia, frequent infections. may resemble atopic dermatitis
- Niacin deficiency / Pellagra – 3D's: dermatitis (casal's necklace), diarrhea, dementia
- Scurvy: Vit C deficiency due to Collagen defect, follicular hyperkeratotic plaques, corkscrew hairs, gingivitis, anemia

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